Ensemble Margin Learning for Noise-Robust Classification of Hyperspectral Remote Sensing Datasets

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***Abstract*—While deep learning models demonstrate remarkable generalization capabilities, their performance remains critically dependent on the quality of training data. Class noise significantly exacerbates overfitting, particularly problematic in hyperspectral images where mislabeling frequently occurs. To address this challenge, a unified margin-based noise filtering framework employing rotation forest (RoF) as its core is proposed, which integrates flexible strategies for deletion and correction. This ensemble approach effectively decouples feature correlations in hyperspectral data while precisely identifying mislabeled samples. Experiments demonstrate the effectiveness of our proposal in performing mapping of land covers using a common deep learning classifier, convolutional neural network (CNN). Generalization performance is measured using overall accuracy (OA) and the hardest-class accuracy as well as McNemar's test. Comparative analyses involve two traditional well-established ensemble-based class noise filters, majority vote and consensus vote filters. The experiments are conducted on six public hyperspectral remote sensing datasets (e.g., Indian Pines, Pavia University, and others), providing a comprehensive evaluation across different environments. Compared to ensemble-based voting-driven filters, on original datasets, a 1.01% increase in OA and 6.02% in the hardest-class accuracy is achieved; meanwhile, on artificially corrupted datasets, a 1.09% increase in OA and 9.72% in the hardest-class accuracy is achieved, and better statistical significance in McNemar's test is obtained.**

***Index Terms*—Deep learning, ensemble classifier, margin, mislabeling, remote sensing.**

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I. Introduction

CLASSIFICATION plays a pivotal role in processing hyperspectral images, as its resulting maps can be exploited in various applications [1], such as agricultural production [2], military reconnaissance [3], oil spill detection [4], environmental monitoring [5], and others [6]. Our focus here is on supervised classification, trained with a significant amount of labeled instances, as opposed to semi-supervised or unsupervised classification methods. As mentioned in [7], the resulting classification accuracy is primarily influenced by the quality of the training data. However, the real-world data often suffer from class noise [8].

Class noise can be categorized into three types [9], [10] based on their relationship with class membership and attribute values. The first type is labeling errors. In the real world, labeling training instances is both costly and subjective [11], [12], requiring significant human effort and domain expertise. This subjectivity often leads to inconsistencies and errors in the labeled data. Additionally, imprecise geolocation [7] further complicates accurate label annotation. In addition to labeling errors, attributes can also be subject to noise. As discussed in [13] and [14], attribute noise is not randomly distributed based on pixel location or class membership in hyperspectral images, and is often considered as mislabeled data. Adding noise to an attribute is equivalent to shifting a given instance in attribute space, possibly to a location where a different label is more appropriate [15]. Sometimes, even outliers and instances from rare classes are regarded as noisy samples [16], [17]. In this study, consistent with the existing literature, we assume a fixed and known rate of class noise.

Over the last decades, many hyperspectral images (HSIs) classification methods have been proposed. In the beginning, a wide spectrum of traditional machine learning algorithms, including support vector machine (SVM) [18], sparse representation [19], [20], [21], random forests (RFs) [22], [23], and kernel-based approaches [24], [25], have become instrumental in various HSI applications. With the emergence of deep learning, various architectures, including autoencoders [26], recurrent neural networks (RNNs) [27], and convolutional neural networks (CNNs) [28], [29], have been extensively employed in this domain. Due to their robust feature learning capabilities, these methods have achieved state-of-the-art performance. More recently, Vision Transformers (ViTs) [30] have emerged as a promising alternative. Unlike CNNs, ViTs excel at capturing long-range dependencies, enabling the utilization of

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global contextual information to further enhance classification accuracy [31], [32].

Despite the demonstrated effectiveness of the aforementioned methods in HSI classification, they heavily rely on the premise that the provided manual labels are entirely accurate. However, the ground-truth labels in HSI are not entirely accurate [33]. Due to the high dimensionality of HSI data, the issue of label noise is particularly challenging to address [34]. This is not only because high dimensionality induces annotation errors, but more fundamentally because the physical characteristics and acquisition complexities of HSI make the very definition of a correct label inherently ambiguous. Furthermore, the nature of high-dimensional data hinders the algorithm's ability to distinguish between noise and hard samples [35].

In accordance with the findings presented in [10], existing approaches for addressing the issue of mislabeling can be broadly categorized into two groups: one involves adjusting conventional techniques to accommodate label noise, while the other needs incorporating a cleansing preprocessing step. The key theme within the first category revolves around loss functions. Various loss functions exhibit sensitivity to label noise, and methods have been proposed to replace them with more robust alternatives. For instance, in [36], the Adaboost (ADB) algorithm is robustified by incorporating modifications derived from a more robust loss function. Similarly, in [37], the same goal is pursued by altering the ADB weights to ensure accurate classification of nonnoisy samples. Certain multiclass classifiers employ a combination of competing binary classifiers in one-versus-all and one-versus-one schemes. It is reported in [38] that the one-versus-one scheme offers greater resilience to noise. Driven by recent strides in deep learning, the dual-channel residual network (DCRN) [39] was designed to counteract label noise in HSI classification by synthesizing intermediate spatial and spectral data. Nevertheless, despite the extensive literature on this subject, our objective is to develop a technique that can be easily integrated with any classification method without requiring an intricate modeling of label noise.

In the second category of methods for handling label noise, unlike unsupervised clustering methods that rely on density peak [40], our approach utilizes a margin-based classifier-induced filtering strategy to identify mislabeled instances. The choice of the induced classifier is quite flexible. These techniques are discussed in [9]. Guided by the context of a specific application using SVM, Ekambaram et al. [41] constrain the support vectors to be selected among noise-free samples, through a manual double-check process. Other techniques assess the likelihood of a given label by examining labels of instances with similar attributes. Reliable samples are those with likelihoods higher than a certain threshold. In [42], this similarity is interpreted as a statistical dependence among attributes conditional on the class. However, as mentioned in [43], similarity cannot be reliably estimated by directly using Euclidean distance in the attribute space, as in k-nearest neighbors (KNNs), due to the high number of attributes in hyperspectral images, which leads to the "curse of dimensionality" or the Hughes phenomenon. In high-dimensional space, the distances between sample points tend to become uniform, leading to a loss of discriminative power in similarity measurement. Therefore, more sophisticated algorithms, such as those based on statistical models or dimensionality reduction techniques, are often required in hyperspectral data to overcome the effects of the "curse of dimensionality" and improve the performance of classification and regression tasks.

Pursuing the same objective, assessing likelihoods while mitigating the Hughes phenomenon, the utilization of ensemble classifiers as detailed in [44] and exemplified by the rotation forest (RoF) classifier has been opted for. This choice provides a versatile framework and is generally less computationally intensive. These ensemble classifiers typically consist of independent decision trees and a majority voting mechanism. Although they inherently offer some robustness to label noise, it has been observed in both [11] and our experiments that their robustness improves when the training sets are filtered. Brodley and Friedl [11] describe two classical learners (CLs); they delete samples for which base learners have no consensus or for which most base learners do not predict the correct label. The first is the consensus filtering technique, often called the unanimity filter, denoted as CL with deletion based on consensus (CLDC). The second is the classical majority filtering technique, denoted as CL with deletion based on majority (CLDM). Complementary to these deletion strategies, we also introduce their correction counterparts: CL with correction based on consensus (CLCC) and CL with correction based on majority (CLCM). All four strategies are included in our comparative analysis.

The consensus filtering technique, often called the unanimity filter, is labeled as CLDC for clarity and tends to reduce type one errors (misidentifying a correct sample as mislabeled) while increasing type two errors (misidentifying a noisy sample as correctly labeled). The classical majority filtering technique, denoted as CLDM, has a similar effect on both types of errors and is preferable when a sufficient number of samples are available. Both of these ensemble filtering techniques are part of our comparative analysis.

In some instances, as demonstrated by the following three examples (but not in our experiments), noise correction improves performance compared to merely removing noise from the datasets. This phenomenon was observed in [42] when correcting only the most probable mislabeled samples. In [45], a heterogeneous ensemble combining four different classifiers is employed to detect and rectify incorrectly labeled samples from training sets, resulting in notably enhanced accuracy, despite an excessively high number of labels identified as noise. In [7], the feature space is first reduced (by averaging nearby wavelength intensities). For each sample, a label likelihood is calculated based on its similarity to other samples sharing the same label. The label is modified when a single label stands out as significantly more likely than all others, with proximity being defined as a quadratic form that incorporates an estimated covariance matrix. It is also proposed that the samples identified by CLDC and CLDM be corrected instead of being deleted. These modified techniques are referred to as CLCC and CLCM, and given their

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conceptual similarity to our proposed correction technique, they are included in our comparative analysis.

As described in [46] and [47], the margin is usually defined in a two-class context, as the distance between training samples and a separating hyperplane. It is considered in the literature as an efficient tool to improve generalization (see [46] recalling a margin-dependent bound on the probabilistic generalization error and [48] for a more involved probabilistic study). It is also defined in the context of ensemble classifiers in a multiclass and in an unsupervised context, as the difference between the estimated likelihoods of the most likely label and the second most likely. It is then called the ensemble margin. It is used in [49] to analyze RF theoretically, in [7] [see (9)] to decide whether an instance should be relabeled, in [50] to show only the most reliable samples, and in [51] to provide the training set with less unreliable samples. With a similar objective, the ensemble margin is also defined in a supervised context in [52] and [53]: it is the difference between the likelihood of the true label and that of all other labels. This last definition is the one used in our proposal.

This work presents an ensemble learning (EL) filter that filters noisy training sets. The proposed EL framework comprises four variants: ensemble learning with deletion based on single-step (ELDS), ensemble learning with deletion based on iterative (ELDI), ensemble learning with correction based on single-step (ELCS), and ensemble learning with correction based on iterative (ELCI), distinguished by their specific combination of handling strategy (deletion or correction) and processing mode (single step or iterative). Those increased performances are first compared with those obtained when induced with filtered datasets by CLDM, CLCM CLDC, and CLCC. All these classification performances are tested on hyperspectral images. The main contributions of this article are listed as follows.

1) A novel class noise ranking function, called ensemble margin-based noise ranking (EMRank), which is based on the ensemble margin, is introduced. The EMRank function ranks samples based on their likelihood of being noisy by evaluating the decision boundaries of the base classifiers in the ensemble. This helps to identify and isolate noisy samples effectively.
2) A maximum performance criterion (MPC) that utilizes a validation set, along with a noise-sensitive training technique, is proposed. By training a noise-sensitive classifier, its performance is evaluated on a specific validation set, and then, the best-performing training set is selected from multiple options.
3) An iterative process that combines both the identification of noisy samples using the EMRank function and the filtering of those samples in a single task is presented. This process allows the above actions to complement each other, progressively enhancing the algorithm's ability to handle noisy datasets.

II. Related Work

A. Random Forest

As an ensemble classifier, RF provides information on the prediction confidence as in [54]; it is successful when dealing with hyperspectral data [43] and large databases [55]. Applied to noisy training datasets, it exhibits robustness as explained in [56].

The RF algorithm, introduced by Breiman [49], is depicted in Algorithm 1. The following classical notations are being used here.

1) $\mathbf{x}_n$ is the nth column feature vector.
2) $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N]^T$ is a feature vector.
3) C is the number of classes.
4) $y \in \{1, \ldots, C\}$ is a label.
5) $Y = [y_1, \ldots, y_N]^T$ is a column label vector.
6) T is the number of iterations.
7) t is an iteration counter.

Algorithm 1 RF

Require: $(\mathbf{X}, Y)$, T, RANDOMIZE, TRAIN_DT
Ensure: $\mathbf{g}$ and G
1: **for** t=1:T **do**
2: $(\mathbf{X}_t, Y_t) = \text{RANDOMIZE}(\mathbf{X}, Y)$
3: $g_t = \text{TRAIN_DT}(\mathbf{X}_t, Y_t)$
4: **end for**
5: $\mathbf{g} = [\mathbf{g}_1 \ldots \mathbf{g}_T]$
6: $G = \text{ENSEMBLE}[(1, g_1), \ldots, (1, g_T)]$

RF consists of processing T independent iterations. In each iteration, a training set is drawn with two actions designed to introduce diversity among training sets. First, an equal number of samples is randomly selected with replacement (a technique also called *bootstrap sampling*), which is denoted as RANDOMIZE in Algorithm 1. Second, a random set of columns is removed, and the number of remaining columns is generally set as the logarithm or the square root of the number of features. These two actions are denoted as $(\mathbf{X}_t, Y_t) = \text{RANDOMIZE}(\mathbf{X}, Y)$. $(\mathbf{X}_t, Y_t)$ induces a decision tree, $g_t = \text{TRAIN_DT}(\mathbf{X}_t, Y_t)$ according to the classification and regression tree (CART) methodology, as proposed in [57].

RF yields T decision trees, denoted as $\mathbf{g} = [g_1, \ldots, g_T]$. They are used to compute NOISE. $\mathbf{g}$ is also used to build an ensemble classifier

$$g = \text{ENSEMBLE}\left[(1, g_1), \ldots, (1, g_T)\right] \tag{1}$$

with

$$\text{ENSEMBLE}\left[g_1, \ldots, g_T\right](\mathbf{x}_n) = \arg\max_{c \in \{1,\ldots,C\}} \sum_{t=1}^{T} \delta\left(g_t(\mathbf{x}_n) = c\right). \tag{2}$$

$\pi \mapsto \delta(\pi)$ is a function mapping any predicate π into 1 or 0 depending on whether π is true or not. RF predictions are decided in accordance with a majority voting, VOTE, counting the decision trees predicting a specific label for a given sample

$$\text{VOTE}(\mathbf{g}, \mathbf{x}_n, c) = \sum_{t=1}^{T} \delta\left(g_t(\mathbf{x}_n) = c\right). \tag{3}$$

Equation (2) can now be written as

$$G(\mathbf{x}_n) = \arg\max_{c \in \{1,\ldots,C\}} \text{VOTE}(\mathbf{g}, \mathbf{x}_n, c). \tag{4}$$

For our purpose, RF is used to get a set of decision trees $\mathbf{g}$; this process is denoted as

$$\mathbf{g} = \text{TRAIN_SDT}((\mathbf{X}, Y), T). \tag{5}$$

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B. Rotation Forest

The RoF, drawing upon the idea of RF, is one of the ensemble methods [58]. The basic concept of RoF is to randomly divide the original feature set into multiple subsets. Similar to the first step of RF, a bootstrap sample is selected from 75% of all training instances. After that, principal component analysis (PCA) is applied on subsets to achieve a rotated matrix. In this experiment, only PCA technology is used to rotate features in hyperspectral images; all eigenvectors corresponding to the eigenvalues of the covariance matrix are selected. This feature extraction technique generates a rotation matrix that allows the original image to be projected into different coordinate systems, thereby reducing the correlation between hyperspectral features and improving the diversity of the base classifier. RoF typically has an advantage in high-dimensional datasets, such as hyperspectral remote sensing images; the introduction of feature subsets and feature transformations helps address the curse of dimensionality. It can handle datasets with a large number of features without the requirement of excessive feature selection or dimensionality reduction steps. In this article, RoF is used to create spatially independent datasets and train base classifiers.

Algorithm 2 RoF

Require: $(\mathbf{X}, Y)$, T, K, SPLITFEATURE, BOOTS, PCA, REARRANGE, TRAIN_DT
Ensure: $\mathbf{g}$ and G
for t=1: T **do**
  for j=1:K **do**
    $\mathbf{X}_{i,j}$ = SPLITFEATURE($\mathbf{X}$)
    $\mathbf{X}'_{i,j}$ = BOOTS($\mathbf{X}_{i,j}$)
    $\mathbf{R}_{i,j}$ = PCA($\mathbf{X}'_{i,j}$)
    $\mathbf{R}_a$ = REARRANGE($\mathbf{R}_{i,j}$)
  end for
  g_t = TRAIN_DT($\mathbf{X} * \mathbf{R}_a, Y$)
end for
$\mathbf{g} = [\mathbf{g}_1 \ldots \mathbf{g}_T]$
G = ENSEMBLE$[(1, g_1), \ldots, (1, g_T)]$

In Algorithm 2, the usage of parameters $(\mathbf{X}, Y)$, T, TRAIN_DT, $\mathbf{g}$, and G is the same as in RF. Before each iteration, the training set is drawn with actions designed to introduce diversity among the base classifiers. First, the feature set is randomly partitioned into K disjoint subsets. This process is formally denoted as SPLITFEATURE. Then, similar to the first step of RANDOMIZE in Algorithm 1, a bootstrap sample is selected from 75% of the number of instances in $\mathbf{X}'_{i,j}$. This step is placed in the function BOOTS, and the result $\mathbf{X}'_{i,j}$ is obtained from this function. After that, PCA is applied on $\mathbf{X}'_{i,j}$ to achieve a rotated matrix $\mathbf{R}_a$. PCA is a traditional method of reducing features. In this experiment, only PCA technology is used to rotate features in hyperspectral images, where all eigenvectors corresponding to the eigenvalues of the covariance matrix are selected. At last, the construction of $\mathbf{R}_a$ is executed by the REARRANGE function, which reorders the columns of all $\mathbf{R}_{i,j}$ to align with the original feature sequence in $\mathbf{X}$.

C. Majority and Consensus Voting Filter

In EL, the majority voting filter is a fundamental decision fusion strategy, commonly used in hard voting ensembles. In this method, multiple base classifiers independently predict the label for the same sample, and the final class is determined by the category that receives the most votes [59]. This approach assumes that all learners are equally reliable. While majority voting effectively reduces the impact of errors from individual classifiers and enhances the model's overall generalization performance, it does not utilize the contextual relationships among predictions. As a result, its effectiveness may be limited in complex or noisy environments.

To address these limitations, a more stringent variant called the consensus voting filter has been developed. The core idea of the consensus voting filter is that a decision is made only when the majority of models unanimously agree on the class of a given sample. In other words, this filter not only counts which class receives the most votes, but also requires that a sufficient level of agreement or consensus is reached among the classifiers [60]. Unlike the traditional majority voting, which only requires a simple majority, consensus voting imposes a higher threshold for agreement. In this article, since the filter is used to determine whether a sample is noisy, the consensus threshold is set to 100%.

D. Convolutional Neural Networks

Deep CNNs stand out as a popular choice in deep learning, particularly within the domain of remote sensing [61]. A typical CNN framework comprises three crucial elements: a convolutional layer, a nonlinear mapping layer, and a pooling operation layer. The creation of a deep CNN involves stacking multiple convolutional and pooling layers. Convolutional operations capture features such as structure, color, and shape from input images, representing them as matrices. These matrices offer abstract attributes of the inherent information in the input data, enabling the neural network to directly acquire high-level features.

III. Proposed Method

A. Description of EMRank

In designing this class noise ranking metric, denoted EMRank, the aim is to find the labeled samples most likely to be incorrectly labeled. EMRank is based on a supervised ensemble margin denoted as MARGIN, which uses $\mathbf{g}$, and is defined in [53]

$$\text{MARGIN}(\mathbf{g}, \mathbf{x}_n, y) = \frac{1}{T}\left(\text{VOTE}(\mathbf{g}, \mathbf{x}_n, y) - \sum_{c \neq y} \text{VOTE}(\mathbf{g}, \mathbf{x}_n, c)\right). \quad (6)$$

As shown in [52], this metric is bounded

$$|\text{MARGIN}(\mathbf{g}, \mathbf{x}_n, y)| \leq 1. \quad (7)$$

An attempt is made to select from the training set, the samples whose labels are the most unlikely. And

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$-\text{MARGIN}(\mathbf{g}, \mathbf{x}_n, y)$ is directly related to the number of decision trees in $\mathbf{g}$ not predicting label y for $\mathbf{x}_n$

$$\begin{aligned} &-\text{MARGIN}(\mathbf{g}, \mathbf{x}_n, y) \\ &= -\frac{1}{T}\text{VOTE}(\mathbf{g}, \mathbf{x}_n, y) + \frac{1}{T}\sum_{c\neq y}\text{VOTE}(\mathbf{g}, \mathbf{x}_n, c) \\ &= \frac{1}{T}\left(-T + \sum_{c\neq y}\text{VOTE}(\mathbf{g}, \mathbf{x}_n, c)\right) + \frac{1}{T}\sum_{c\neq y}\text{VOTE}(\mathbf{g}, \mathbf{x}_n, c) \\ &= \frac{2}{T}\sum_{c\neq y}\text{VOTE}(\mathbf{g}, \mathbf{x}_n, c) - 1. \end{aligned} \tag{8}$$

A related metric denoted $EMRank(\mathbf{g}, \mathbf{x}_n, y)$ was defined, which indicates how likely it is that the label y is wrong. If G happens to predict the true label y, it is equal to 0 and if not, it is defined as the opposite of the margin

$$\begin{aligned} &EMRank(\mathbf{g}, (\mathbf{x}_n, y)) \\ &= \begin{cases} 0, & \text{if } G(\mathbf{x}_n) = y \\ -\text{MARGIN}(\mathbf{g}, \mathbf{x}_n, y), & \text{if } G(\mathbf{x}_n) \neq y. \end{cases} \end{aligned} \tag{9}$$

Algorithm 3 $\mathcal{S}' = \text{SELECT}(\mathbf{g}, \mathcal{S}, a)$

Require: $\mathbf{g}$, $\mathcal{S}$, a
Ensure: $\mathcal{S}'$
1: **for** $n = 1 : \#\mathcal{S}$ **do**
2: Read the nth sample $(\mathbf{x}_n, y_n)$ of $\mathcal{S}$.
3: Compute $r_n = EMRank(\mathbf{g}, (\mathbf{x}_n, y_n))$
4: **end for**
5: Compute a permutation φ sorting $r_1 \dots r_N$
6: Select the a first samples: $\mathcal{S}' = \{(\mathbf{x}_{\varphi(1)}, y_{\varphi(1)}), \dots, (\mathbf{x}_{\varphi(a)}, y_{\varphi(a)})\}$

The selecting procedure called SELECT uses EMRank: given an array of decision trees $\mathbf{g}$, a training set now denoted as $\mathcal{S}$, and a number of samples a, it yields a set containing the a-samples whose labels are the most unlikely. It is described in Algorithm 3 and used in Sections III-C and III-D.

B. Maximum Performance Criterion

A deep learning classifier, the CNN, is here described, and it is used to estimate the performance of $\mathcal{S} = (\mathbf{X}, Y)$, a filtered dataset. The resulting classifier is denoted as follows:

$$H = \text{TRAIN_CNN}(\mathcal{S}, T). \tag{10}$$

EMRank defined in (9) provides ranking information, not the amount of samples that should be corrected or removed. It is the purpose of the MPC to select among a number of training sets, the best one. Through training a noise-sensitive classifier, its performance is assessed with the validation set, and then, the best-performing set is selected.

Given an available training set $\mathcal{S}$, a validation set $\mathcal{S}_{\mathbb{V}}$, and a reduced training set $\mathcal{S}_{\mathbb{T}}$ are computed using a SPLIT-function, by randomly moving all rows of $\mathcal{S}$, (those of the feature matrix and the corresponding components of the label vector) to either $\mathcal{S}_{\mathbb{T}}$ or $\mathcal{S}_{\mathbb{V}}$

$$[\mathcal{S}_{\mathbb{T}}, \mathcal{S}_{\mathbb{V}}] = \text{SPLIT}(\mathcal{S}). \tag{11}$$

MPC is described in Algorithm 4, and it consists of the following tasks. The inputs are $\mathcal{S}_{V_m}$ and a certain amount of training sets denoted as $\mathcal{S}_{T_m}$ obtained with information provided only by $\mathcal{S}_{\mathbb{V}}$ and $\mathcal{S}_{\mathbb{T}}$. m noise-sensitive classifiers are induced in accordance with the algorithm CNN, one for each training set

$$\mathcal{H}_m = \text{TRAIN_CNN}(\mathcal{S}_{T_m}, T). \tag{12}$$

The overall accuracy (OA) of all m induced classifiers is computed on $\mathcal{S}_{V_m}$

$$\text{TEST}(\mathcal{H}_m, \mathcal{S}_{V_m}) = \frac{1}{\#\mathcal{S}_{V_m}}\sum_{(\mathbf{x},y)\in\mathcal{S}_{V_m}}\delta(\mathcal{H}_m(\mathbf{x}) = y) \tag{13}$$

where $\#\mathcal{S}_{V_m}$ is the size of $\mathcal{S}_{V_m}$. Finally, the MPC consists of selecting the best-performing candidate

$$\text{MPC}(\mathcal{S}_{T_m}, \mathcal{S}_{V_m}) = \arg\max_m \text{TEST}(\mathcal{H}_m, \mathcal{S}_{V_m}). \tag{14}$$

Algorithm 4 MPC

Require: $\mathcal{S}_{V_1}, \dots, \mathcal{S}_{V_M}$, $\mathcal{S}_{T_1}, \dots, \mathcal{S}_{T_M}$, TRAIN_CNN, T
Ensure: $\widehat{m}$
1: **for** m=1: M **do**
2: $\mathcal{H}_m = \text{TRAIN_CNN}(\mathcal{S}_{T_m}, T)$
3: $\xi_m = \text{TEST}(\mathcal{H}_m, \mathcal{S}_{V_m})$
4: **end for**
5: $\widehat{m} = \arg\max_m \xi_m$

Algorithm 5 $\mathcal{S}'' = \text{DELETE}(\mathcal{S}, \mathcal{S}')$

Require: $\mathcal{S}$ and $\mathcal{S}'$
Ensure: $\mathcal{S}''$
1: $\mathcal{S}'' = \mathcal{S}$
2: **for** $n = 1 : \#\mathcal{S}'$ **do**
3: Read $(\mathbf{x}_n, y)$ from $\mathcal{S}'$
4: **if** $(\mathbf{x}_n, y_n) \in \mathcal{S}''$ **then**
5: Delete $(\mathbf{x}_n, y_n)$ from $\mathcal{S}''$
6: **end if**
7: **end for**

C. Description of the ELCS and ELDS Algorithms

The complete description of the algorithm still needs to make explicit the use of SELECT to build the M filtered datasets using EMRank. M amounts of labels denoted as $a_1, \dots, a_M$ and ranging from 0 to $2/5\#\mathcal{S}$ with an approximate step of $1/100\#\mathcal{S}$ are considered

$$a_m = \text{ROUND}\left(\frac{m\#\mathcal{S}}{100}\right) \text{ for } m \in \{0, \dots, 40\} \tag{15}$$

where ROUND maps reals into their closest integer, and $\#\mathcal{S}$ is the number of samples in the training set $\mathcal{S}$. The M filtered datasets are $\text{SELECT}(\mathbf{g}, \mathcal{S}_{\mathbb{T}}, a_m)$ for $m \in \{1, \dots, M\}$.

Algorithms 5 and 6 are used to either delete or correct these M filtered datasets. These two operators transform a given training set $\mathcal{S}$ and a given selected set $\mathcal{S}'$ into a new training set

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$\mathcal{S}''$, where either the selected samples are deleted or corrected according to a given classifier.

Algorithm 6 $\mathcal{S}''$ = CORRECT$(\mathcal{S}, \mathcal{G}, \mathcal{S}')$

Require: $\mathcal{S}$, $\mathcal{S}'$ and $\mathcal{G}$
Ensure: $\mathcal{S}''$
$\mathcal{S}'' = \mathcal{S}$
for $n = 1 : \#\mathcal{S}'$ **do**
  Read $(\mathbf{x}_n, y)$ from $\mathcal{S}'$
  if $(\mathbf{x}_n, y_n) \in \mathcal{S}''$ **then**
    Replace $(\mathbf{x}_n, y_n)$ in $\mathcal{S}''$ by $(\mathbf{x}_n, \mathcal{G}(\mathbf{x}_n))$
  end if
end for

Algorithm 7 TRAIN_ELDS and TRAIN_ELCS

Require: $\mathcal{S}$, T, TRAIN_SDT, TRAIN_CNN, option $\in \left\{ \begin{matrix} \text{D}, \\ \text{C} \end{matrix} \right\}$
Ensure: $\widehat{\mathcal{S}}$, $\widehat{\mathcal{H}}$
$(\mathcal{S}_{\mathbb{T}}, \mathcal{S}_{\mathbb{V}})$ = SPLIT$(\mathcal{S})$
$\mathbf{g}$ = TRAIN_SDT$(\mathcal{S}, T)$
$\mathcal{S}'_{T_0} = \mathcal{S}_{\mathbb{T}}$
$\mathcal{S}'_{V_0} = \mathcal{S}_{\mathbb{V}}$
for $m = 1 : M$ **do**
  a_{T_m} = ROUND $\left(\frac{m\#\mathcal{S}_{\mathbb{T}}}{100}\right)$
  a_{V_m} = ROUND $\left(\frac{m\#\mathcal{S}_{\mathbb{V}}}{100}\right)$
  $\mathcal{S}_{T_m}$ = SELECT$(\mathbf{g}, \mathcal{S}_{\mathbb{T}}, a_{T_m})$
  $\mathcal{S}_{V_m}$ = SELECT$(\mathbf{g}, \mathcal{S}_{\mathbb{V}}, a_{V_m})$
  if option = delete **then**
    $\mathcal{S}'_{T_m}$ = DELETE$(\mathcal{S}_{\mathbb{T}}, \mathcal{S}_{T_m})$
    $\mathcal{S}'_{V_m}$ = DELETE$(\mathcal{S}_{\mathbb{V}}, \mathcal{S}_{V_m})$
  else
    $\mathcal{S}'_{T_m}$ = CORRECT$(\mathcal{S}_{\mathbb{T}}, \mathcal{G}, \mathcal{S}_{T_m})$
    $\mathcal{S}'_{V_m}$ = CORRECT$(\mathcal{S}_{\mathbb{V}}, \mathcal{G}, \mathcal{S}_{V_m})$
  end if
  Save$(\mathcal{S}'_{T_m}, \mathcal{S}'_{V_m})$
end for
$\widehat{m}$ = MPC$(\mathcal{S}'_{V_1}, \ldots, \mathcal{S}'_{V_M}, \mathcal{S}'_{T_1}, \ldots, \mathcal{S}'_{T_M}, \text{TRAIN_CNN}, T)$
$\widehat{\mathcal{S}} = \mathcal{S}'_{\widehat{m}}$
$\widehat{\mathcal{H}}$ = TRAIN_CNN$(\widehat{\mathcal{S}})$

The different tools already described allow us to introduce ELDS and ELCS, as described in Algorithm 7. Given a training set $\mathcal{S}$, a classifying technique yielding decision trees TRAIN_SDT, and a noise-sensitive classifying technique TRAIN_CNN, these algorithms yield a training set and the induced classifier. The validation set is first created with SPLIT, and some primary information is retrieved from the dataset by inducing decision trees and an RF classifier with TRAIN_SDT. A greedy technique consisting of generating a certain amount of training sets with the most unlikely labels using EMRank and SELECT is used. At last, MPC and the created validation set are used to select the best training set, and TRAIN_CNN is used to induce the corresponding classifier.

Algorithm 8 TRAIN_ELDI and TRAIN_ELCI

Require: $\mathcal{S}$, T, TRAIN_SDT, TRAIN_CNN, option $\in \left\{ \begin{matrix} \text{D}, \\ \text{C} \end{matrix} \right\}$
Ensure: $\widehat{\mathcal{S}}$, $\widehat{\mathcal{H}}$
$(\mathcal{S}_{\mathbb{T}}, \mathcal{S}_{\mathbb{V}})$ = SPLIT$(\mathcal{S})$
$\mathcal{S}'_{T_0} = \mathcal{S}_{\mathbb{T}}$
$\mathcal{S}'_{V_0} = \mathcal{S}_{\mathbb{V}}$
$(\mathbf{g}_0, \mathcal{G}_0)$ = TRAIN_SDT$(\mathcal{S}, T)$
$a_{T_0} = 0$
$a_{V_0} = 0$
for $m = 1 : M$ **do**
  a_{T_m} = ROUND $\left(\frac{m\#\mathcal{S}_{\mathbb{T}}}{100}\right)$
  a_{V_m} = ROUND $\left(\frac{m\#\mathcal{S}_{\mathbb{V}}}{100}\right)$
  $\mathcal{S}_{T_m}$ = SELECT$(\mathbf{g}_{m-1}, \mathcal{S}'_{T_{m-1}}, a_{T_m} - a_{T_{m-1}})$
  $\mathcal{S}_{V_m}$ = SELECT$(\mathbf{g}_{m-1}, \mathcal{S}'_{V_{m-1}}, a_{V_m} - a_{V_{m-1}})$
  if option = delete **then**
    $\mathcal{S}'_{T_m}$ = DELETE$(\mathcal{S}_{\mathbb{T}}, \mathcal{S}_{T_m})$
    $\mathcal{S}'_{V_m}$ = DELETE$(\mathcal{S}_{\mathbb{V}}, \mathcal{S}_{V_m})$
  else
    $\mathcal{S}'_{T_m}$ = CORRECT$(\mathcal{S}_{\mathbb{T}}, \mathcal{G}_{m-1}, \mathcal{S}_{T_m})$
    $\mathcal{S}'_{V_m}$ = CORRECT$(\mathcal{S}_{\mathbb{V}}, \mathcal{G}_{m-1}, \mathcal{S}_{V_m})$
  end if
  Save$(\mathcal{S}'_{T_m}, \mathcal{S}'_{V_m})$
  $\mathcal{S}'_m$ = COMBINE$(\mathcal{S}'_{T_m}, \mathcal{S}'_{V_m})$
  $(\mathbf{g}_m, \mathcal{G}_m)$ = TRAIN_SDT$(\mathcal{S}'_m, T)$
end for
$\widehat{m}$ = MPC$(\mathcal{S}'_{V_1}, \ldots, \mathcal{S}'_{V_M}, \mathcal{S}'_{T_1}, \ldots, \mathcal{S}'_{T_M}, \text{TRAIN_CNN}, T)$
$\widehat{\mathcal{S}} = \mathcal{S}'_{\widehat{m}}$
$\widehat{\mathcal{H}}$ = TRAIN_CNN$(\widehat{\mathcal{S}}, T)$

D. Description of the ELDI and ELCI Algorithms

An expected weakness in ELDS and ELCS is that at iteration m, the selection of samples having the most unlikely labels uses information retrieved from $\mathcal{S}_{\mathbb{T}}$, which includes either some samples that have been deleted or corrected in the previous iterations. As a result, the improvement obtained with the previous iterations is not taken into account in selecting samples. To solve this problem, ELDI and ELCI use the modified training sets by training at each iteration a new set of decision trees and their associated ensemble classifier to better select the samples to be corrected or deleted. They are described in Algorithm 8. The COMBINE function merges two different datasets into one

$$\mathcal{S} = \text{COMBINE}\left[\mathcal{S}_{\mathbb{T}}, \mathcal{S}_{\mathbb{V}}\right]. \tag{16}$$

Therefore, the number of selected samples to be modified is reduced, which is the difference between a_m and a_{m-1}.

The flowcharts of the proposed Algorithms 7 and 8 are displayed separately in Fig. 1.

IV. Experimental Results

A. Experimental Datasets

There are six groups of hyperspectral datasets utilized in this article. Figs. 2–7 illustrate the main object classes of the six hyperspectral datasets. To better validate the performance

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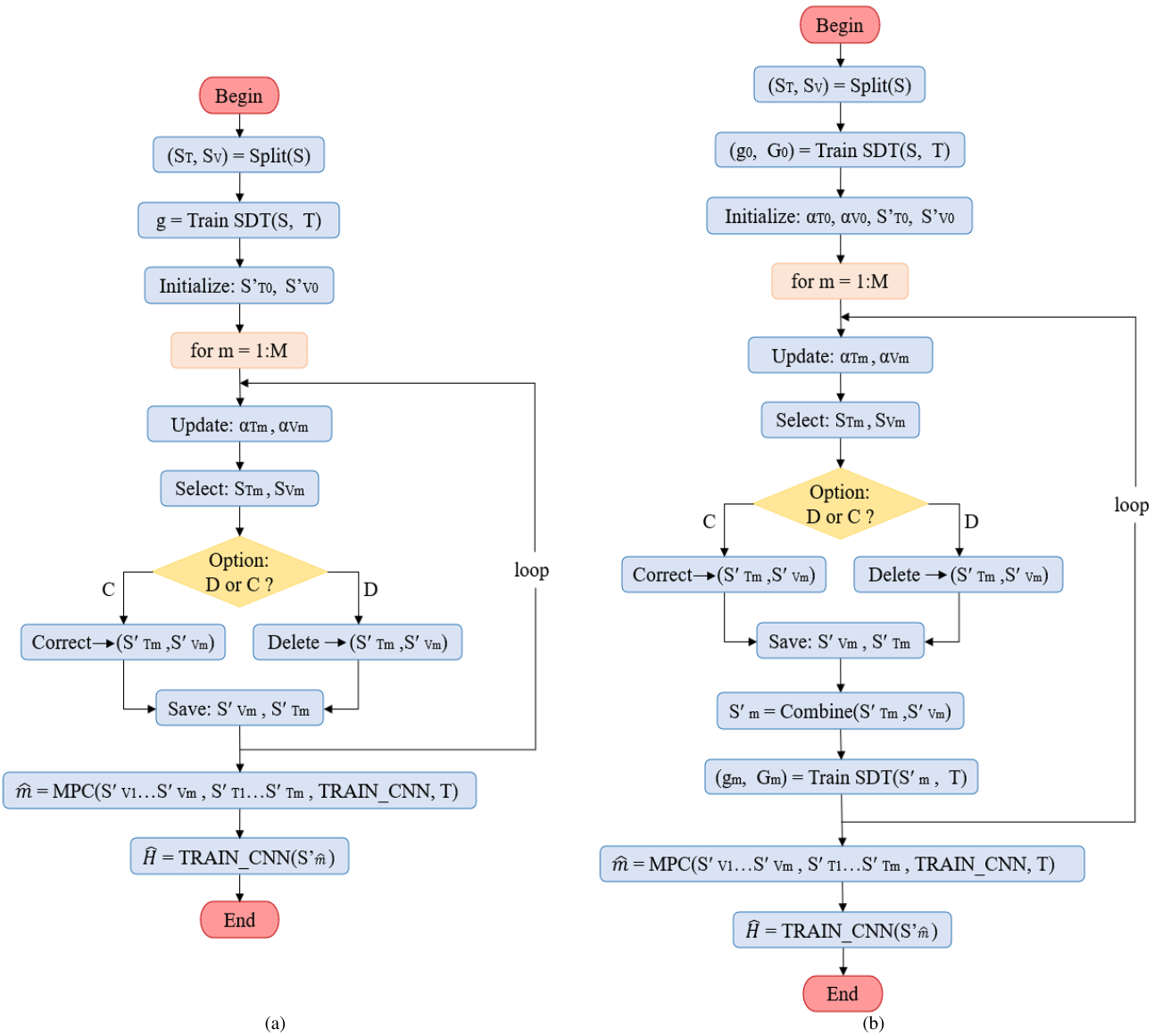


Fig. 1. Flowcharts of the proposed class noise cleaning algorithms. (a) Flowchart of the ELCS and ELDS algorithms. (b) Flowchart of the ELDI and ELCI algorithms.

of the proposed model, multiple experiments were conducted, and the results were averaged.

The scene size of *Indian Pines* was 145×145 with a spatial resolution of 20 m/pixel and 220 spectral bands ranging from 400 to 2500 nm at a 10-nm spectral resolution. The 20 spectral noisy bands were removed, leaving 200 spectral bands for classification.

Pavia University was obtained by German airborne ROSIS sensor in Northern Italy in 2003. The ROSIS sensor could capture spectral bands covering the range of 430–860 nm, with a spatial resolution of 1.3 m/pixel. The number of bands was 103.

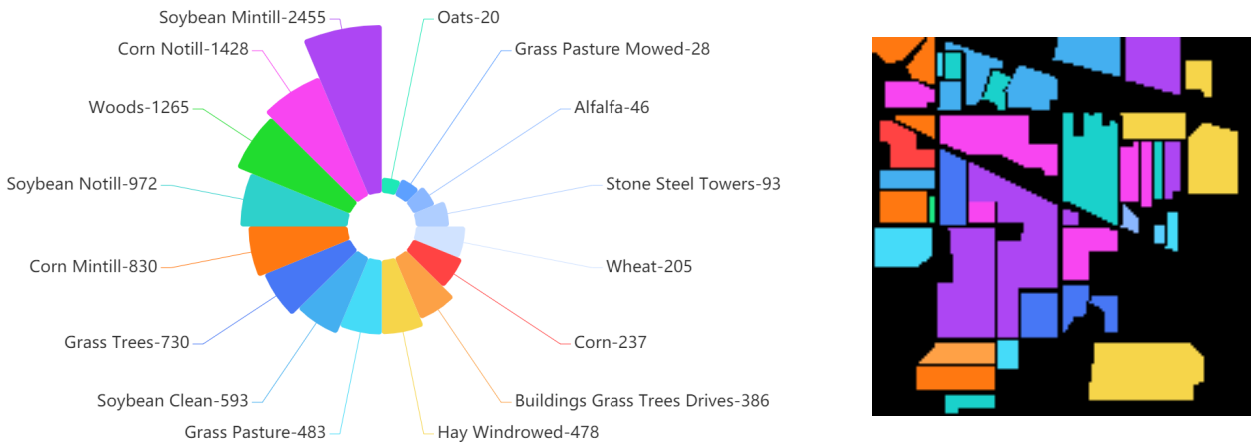
Houston data were acquired by the ITRES CASI-1500 sensor and provided by the 2013 IEEE GRSS Data Fusion Contest. The data had dimensions of 349×1905 and includes 144 bands spanning the spectral range from 364 to 1046 nm.

The *Botswana* dataset was obtained by NASA's EO-1 satellite over the ODeltago delta. The scene was composed of 1476×256 pixels, with a spatial resolution of 30 m/pixel. After removing uncalibrated, water absorption, and noisy bands, it left 145 bands containing 14 categories.

AVIRIS sensor captured data from the *Kennedy Space Center* (KSC) in 224 bands with center wavelengths ranging from 400 to 2500 nm. This dataset contained 224 bands, and the remaining 176 bands were used for analysis after filtering out low signal-to-noise ratio (SNR) bands and removing water absorption.

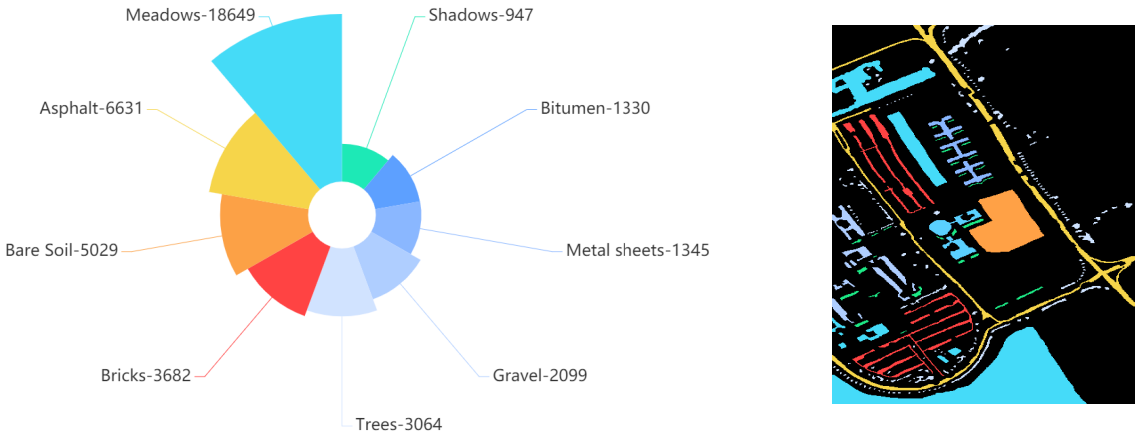
The *Xuzhou* dataset was collected by an airborne HYSPEX hyperspectral camera over the Xuzhou peri-urban site in November 2014. This dataset consisted of 500×260 pixels, with the spatial resolution of 0.73 m/pixel. The number of spectral bands in this experiment was 436.

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(a) (b)

Fig. 2. Visualization of the class distribution in Indian Pines dataset. (a) Pie chart of the number of samples in each class. (b) Ground-truth map of Indian Pines dataset.



(a) (b)

Fig. 3. Visualization of the class distribution in Pavia University dataset. (a) Pie chart of the number of samples in each class. (b) Ground-truth map of Pavia University dataset.

TABLE I
PARAMETERS OF THE DEEP CNN FOR EXPERIMENTS

Data	Window Size	Convolution Activation	Kernel Size	Fully1 Activation	Fully2 Activation	Fully3
Dataset	$1 \times 1 \times N$	$1 \times 1 \times A$ ReLU	B	Fully Connected ReLU	Fully Connected ReLU	C Softmax

B. Experimental Settings

In this experiment, RoF [58] with CART [57] was used as the base classifier to create an ensemble classifier. In RoF, 20 features were taken as a group, rotating the features and reorganizing them with the classic orthogonal transformation method: PCA. RoF was implemented with 100 pruned trees. To prevent rotate forest from overfitting, the parameter "MaxNumSplits" of pruned trees was set to 160. Other parameters of CART were kept to their default values in MATLAB. The CNN parameters used in the experiments are shown in Table I, where N refers to the dimension of the data, A is the vector [25, 10, 5, 1], corresponding to [16, 32, 32, 16] in B, and C is the number of categories of the dataset.

This experiment involved two main parts: one contained direct classification on public hyperspectral images without added class noise, while the other was artificially corrupted with a 20% noise rate. In reality, class noise is typically random and unknown [62]; in order to obtain mislabeling,

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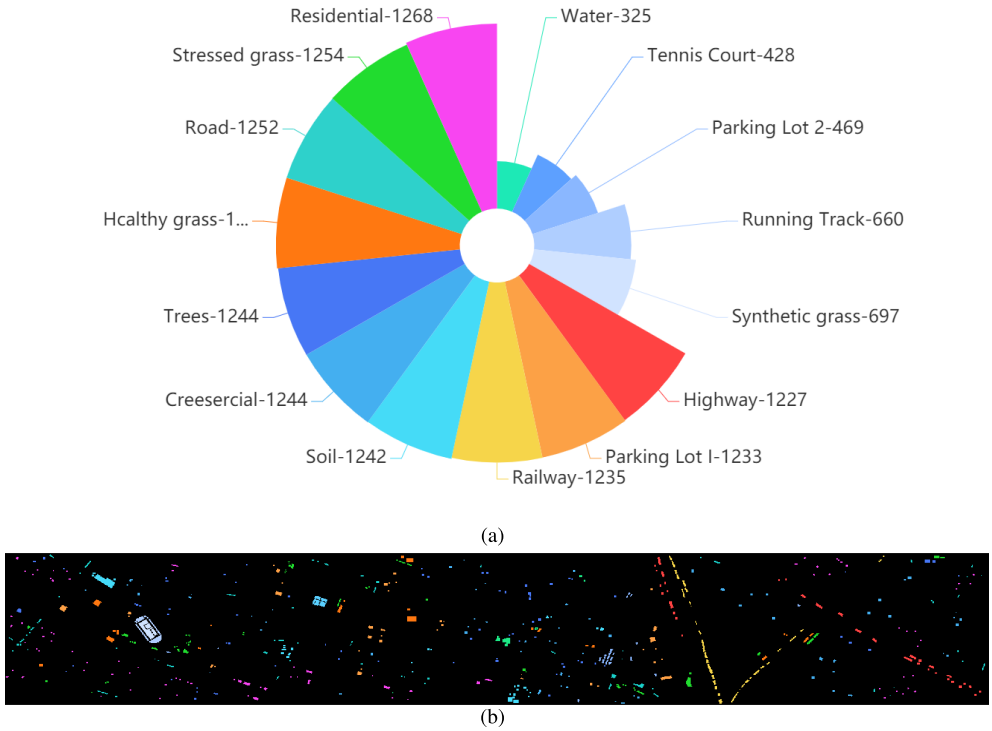


Fig. 4. Visualization of the class distribution in Houston dataset. (a) Pie chart of the number of samples in each class. (b) Ground-truth map of Houston dataset.

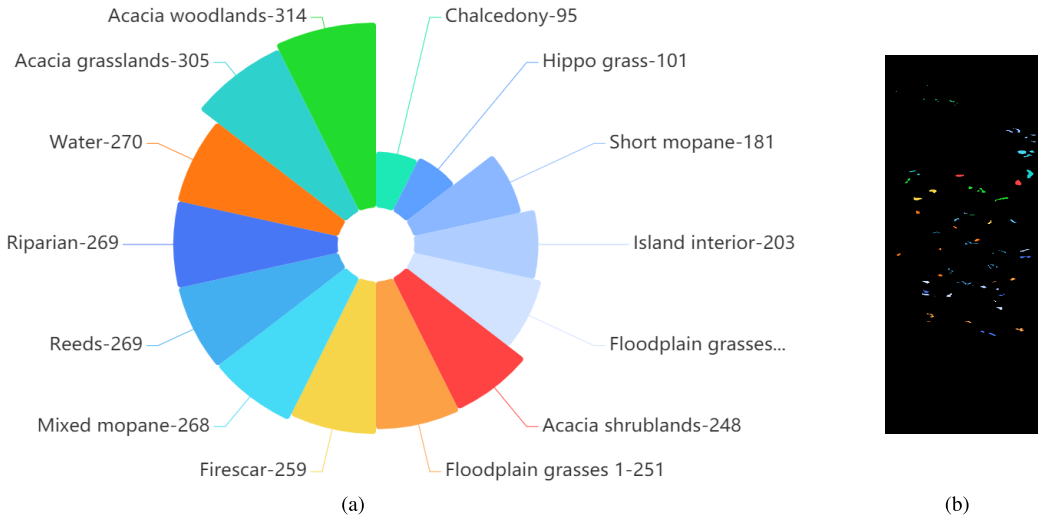


Fig. 5. Visualization of the class distribution in Botswana dataset. (a) Pie chart of the number of samples in each class. (b) Ground-truth map of Botswana dataset.

researchers often replaced labels with other classes [11], [63]. A similar approach was adopted in our experiment. Each algorithm was run five times, and results were averaged to two decimal places, with the best results highlighted in bold in the tables.

All techniques use a validation dataset, which is 33% of the training samples, leaving 67% for training. The test set also comprises 33% of all samples. To ensure fairness and minimize the impact of training sample size on accuracy, the validation set was not merged into the training set for the majority and consensus vote methods, unlike the approach in [64]; instead, it was kept separate to test model robustness.

Evaluation metrics are essential to experiments. Classification performance was assessed with two key metrics: OA and minimum recall per class, which reflects the measure of the accuracy of the hardest class. The latter metric is particularly valuable when dealing with imbalanced data, making it a more robust indicator than OA. Besides, the significance of the differences in classification performance between the margin-based noise filters and other filters is evaluated by McNemar's

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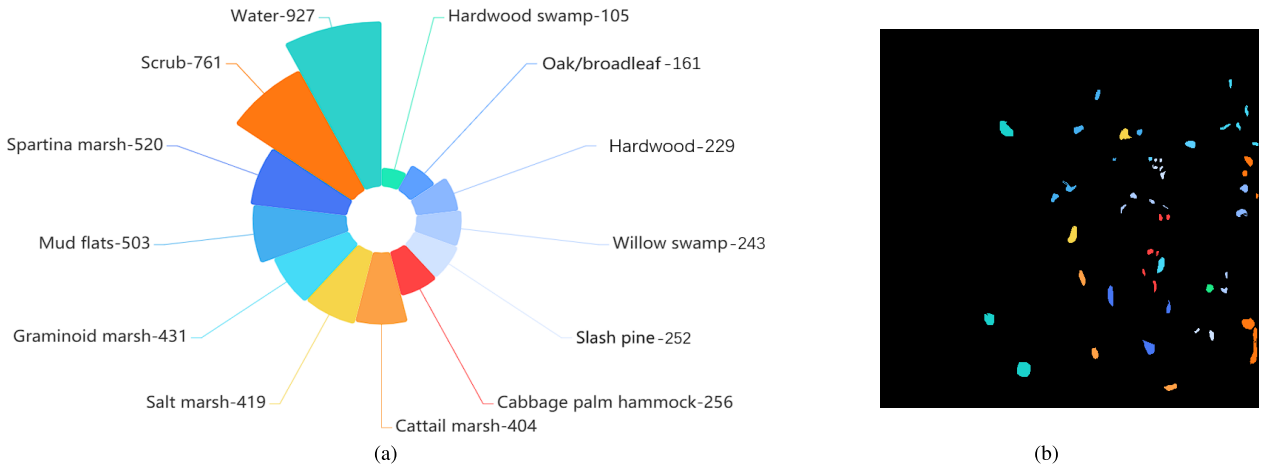


Fig. 6. Visualization of the class distribution in the Kennedy space center dataset. (a) Pie chart of the number of samples in each class. (b) Ground-truth map of the KSC dataset.

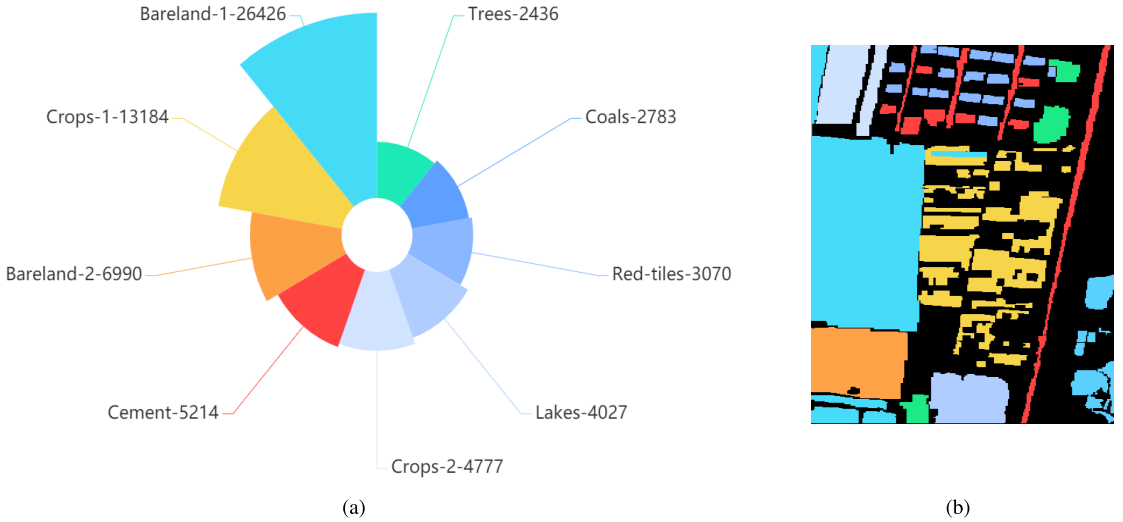


Fig. 7. Visualization of the class distribution in Xuzhou dataset. (a) Pie chart of the number of samples in each class. (b) Ground-truth map of Xuzhou dataset.

test (Z) [65]. McNemar's test calculates Z using the formula $Z = (F_{12} - F_{21})/((F_{12} + F_{21}))^{1/2}$, where F_{12} represents the number of samples correctly classified by filter 1 and misclassified by filter 2, and F_{21} represents the number of samples correctly classified by filter 2 and misclassified by filter 1.

C. Results of Experiments

1) Overall Accuracy: Tables II and III present the OA of CNN when induced with unfiltered and filtered with CLDM, CLDC, ELDS, ELDI, and CLCM, CLCC, ELCS, ELCI.

As shown in Table II, on the original datasets, the primary performance differentiator lies not in the choice between deletion and correction strategies, but rather in the incorporation of the "margin" mechanism. Specifically, the baseline method CNN achieves only 83.16% on complex datasets such as Indian Pines and 91.07% on KSC, ranking sventh and sixth, respectively. In contrast, methods incorporating the margin mechanism (the EL series) consistently occupy the top four positions across almost all datasets. Taking Botswana dataset as an example, the CLDM method (without margin) achieves 89.99%, whereas the ELDS method (with margin) improves to 93.20%. Similarly, on Xuzhou dataset, the margin-based ELDS outperforms the nonmargin CLDC and CLDM. This consistent increase demonstrates the effectiveness of the margin mechanism in enhancing interclass separability. From an overall perspective, the average accuracy of the ELCS method reaches 91.03%, surpassing the raw baseline by 1.4%, further validating the generalization ability of the margin constraint.

In Table III, the performance disparity among algorithms becomes particularly pronounced on challenging datasets. For instance, on Indian Pines, which presents significant classification challenges, the baseline (CNN) accuracy drops to 71.99% (ranking ninth). A similar trend is observed on the KSC dataset, where the raw accuracy is suppressed to 85.69%. Without boundary constraints, label correction may lead to

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TABLE II
OA IN NINE EXPERIMENTS ON ORIGINAL DATASETS

	Raw	Deletion				Correction			
		CLDM	CLDC	Margin		CLCM	CLCC	Margin	
				ELDS	ELDI			ELCS	ELCI
Indian Pines	83.16%(7)	83.03%(8)	83.62%(6)	85.10%(2)	84.37%(4)	84.98%(3)	82.66%(9)	**86.29%**(1)	84.21%(5)
PaviaU	90.99%(7)	90.89%(8)	90.68%(9)	91.82%(2)	91.57%(3)	91.18%(5)	91.32%(4)	91.09%(6)	**91.90%**(1)
Houston	89.95%(7)	90.02%(6)	89.91%(9)	**90.98%**(1)	90.93%(2)	90.07%(5)	89.92%(8)	90.61%(3)	90.38%(4)
Botswana	87.76%(9)	89.99%(8)	90.06%(7)	93.20%(2)	92.33%(4)	90.56%(6)	91.24%(5)	93.17%(3)	**93.32%**(1)
KSC	91.07%(6)	89.42%(9)	90.83%(7)	91.38%(4)	**92.24%**(1)	90.74%(8)	91.27%(5)	91.71%(3)	91.99%(2)
Xuzhou	92.64%(8)	92.87%(7)	93.11%(5)	**93.69%**(1)	93.45%(2)	92.58%(9)	93.03%(6)	93.29%(4)	93.41%(3)
Average	89.63%(8)	89.37%(9)	89.70%(7)	**91.03%**(1)	90.98%(3)	90.02%(5)	89.91%(6)	**91.03%**(1)	90.87%(4)

TABLE III
OA IN NINE EXPERIMENTS ON ARTIFICIALLY CORRUPTED DATASETS (NOISE RATE = 20%)

	Raw	Deletion				Correction			
		CLDM	CLDC	Margin		CLCM	CLCC	Margin	
				ELDS	ELDI			ELCS	ELCI
Indian Pines	71.99%(9)	80.83%(5)	76.42%(7)	81.68%(4)	82.89%(2)	81.97%(3)	73.51%(8)	**83.26%**(1)	80.56%(6)
PaviaU	86.81%(9)	89.77%(5)	88.79%(8)	90.10%(2)	**90.37%**(1)	89.96%(4)	88.93%(7)	90.10%(2)	89.66%(6)
Houston	86.86%(9)	88.31%(7)	88.50%(6)	**90.47%**(1)	89.87%(4)	88.54%(5)	88.29%(8)	90.22%(3)	90.26%(2)
Botswana	83.76%(9)	87.75%(5)	84.89%(8)	88.96%(3)	**89.30%**(1)	86.79%(6)	85.23%(7)	89.16%(2)	88.92%(4)
KSC	85.69%(8)	87.94%(6)	85.35%(9)	88.99%(4)	89.02%(3)	88.99%(4)	85.82%(7)	**89.54%**(1)	89.13%(2)
Xuzhou	91.88%(9)	92.72%(2)	92.48%(7)	92.62%(3)	92.51%(5)	92.51%(5)	92.33%(8)	**93.05%**(1)	92.55%(4)
Average	84.50%(9)	87.89%(6)	86.07%(7)	88.80%(3)	88.99%(2)	88.13%(5)	85.64%(8)	**89.22%**(1)	88.51%(4)

error propagation. For example, the correction method without margin (CLDC) achieves only 76.42% on Indian Pines. In comparison, the ELCS method, which combines the correction strategy with the margin mechanism, reaches an accuracy of 83.26% on Indian Pines. Looking at the average metrics, ELCS achieves 89.22%, which is significantly higher than the nonmargin correction method CLCC and the baseline CNN. This improvement of over 11 percentage points relative to the baseline on Indian Pines, and the nearly 5% improvement on average, highlights the method's robustness in handling complex, noisy regions. Notably, although ELCS yields the best overall performance, deletion-based strategies (such as ELDI and ELDS) achieve the highest scores on the PaviaU and Houston datasets. Specifically, on PaviaU, ELDI achieves 90.37%, slightly outperforming ELCS at 90.10%. This indicates that under certain feature distributions, directly discarding noisy samples may be more conducive to maintaining training data purity than attempting to correct them.

Based on the combined results from Tables II and III, our proposed algorithms exhibit distinct advantages over baselines and traditional ensemble filters: Primarily, the proposed methods consistently dominate the top tier of performance, sweeping ranks 1–4 across both tables. This evidence strongly suggests that the ensemble margin mechanism offers superior accuracy in noise identification compared to conventional majority/consensus voting schemes. Furthermore, algorithmic safety is critical when dealing with original datasets containing unknown noise. In contrast to our margin-based approach, traditional methods such as CLDM suffered from accuracy regression on the KSC dataset and average metrics, leading to the detrimental loss of informative training data. Ultimately, in data-scarce or high-noise environments, the proposed algorithms not only detect noise but also utilize ensemble knowledge to rectify incorrect labels. By correcting rather than discarding samples, the method effectively augments the training set size, a factor that is vital for optimizing deep learning models.

In order to display the classification results of various methods better, Figs. 8–13 show the land cover mapping of original datasets and artificially corrupted datasets (noise rate = 20%) by the methods above.

2) Hardest-Class Accuracy: Similar to Tables II and III, Tables IV and V show the hardest-class accuracy of CNN in original and artificially corrupted datasets when processed with

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Fig. 8. Classification maps of CNN, majority correction and removing, consensus correction and removing, iteration and single-step correction, and iteration and single-step removing, on the original hyperspectral data *Indian Pines*. (a) CNN. (b) CLCM. (c) CLDM. (d) CLCC. (e) CLDC. (f) ELCI. (g) ELCS. (h) ELDI. (i) ELDS.

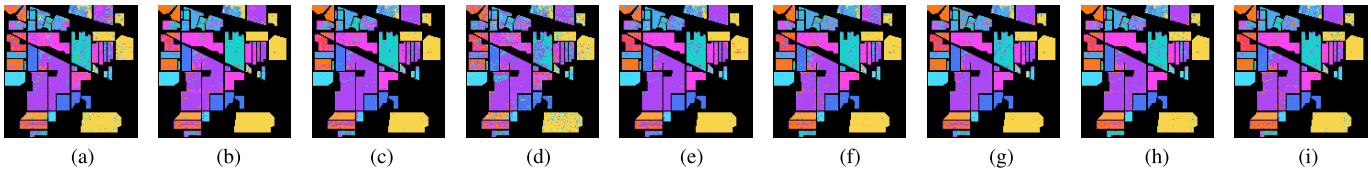


Fig. 9. Classification maps of CNN, majority correction and removing, consensus correction and removing, iteration and single-step correction, and iteration and single-step removing, on artificially corrupted hyperspectral data *Indian Pines* (noise rate = 20%). (a) CNN. (b) CLCM. (c) CLDM. (d) CLCC. (e) CLDC. (f) ELCI. (g) ELCS. (h) ELDI. (i) ELDS.



Fig. 10. Classification maps of CNN, majority correction and removing, consensus correction and removing, iteration and single-step correction, and iteration and single-step removing, on the original hyperspectral data *Pavia University*. (a) CNN. (b) CLCM. (c) CLDM. (d) CLCC. (e) CLDC. (f) ELCI. (g) ELCS. (h) ELDI. (i) ELDS.



Fig. 11. Classification maps of CNN, majority correction and removing, consensus correction and removing, iteration and single-step correction, and iteration and single-step removing, on artificially corrupted hyperspectral data *Pavia University* (noise rate = 20%). (a) CNN. (b) CLCM. (c) CLDM. (d) CLCC. (e) CLDC. (f) ELCI. (g) ELCS. (h) ELDI. (i) ELDS.



Fig. 12. Classification maps of CNN, majority correction and removing, consensus correction and removing, iteration and single-step correction, and iteration and single-step removing, on the original hyperspectral data *Xuzhou*. (a) CNN. (b) CLCM. (c) CLDM. (d) CLCC. (e) CLDC. (f) ELCI. (g) ELCS. (h) ELDI. (i) ELDS.

unfiltered and eight filters used above. In this experiment, the hardest class is defined as the category with the lowest recall in each dataset.

In Table IV, the top-performing algorithm in terms of average metrics, ELDS, outperformed the baseline CNN model by 10.43 percentage points; notably, ELDS achieved an accuracy of 87.59% on Botswana dataset. This advantage is further amplified in Table V, where the best-performing ELDI surpassed the raw baseline by 18.26 percentage points, representing a significant relative improvement of 38.6%. It is particularly worth noting that on Indian Pines dataset, which was severely impacted by noise, the accuracy of CNN plummeted to 30.00%, whereas ELDI maintained a level of 67.50%, demonstrating the exceptional robustness of this class of algorithms in adverse environments. Furthermore, from a holistic perspective, the OA of margin-based algorithms

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(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 13. Classification maps of CNN, majority correction and removing, consensus correction and removing, iteration and single-step correction, and iteration and single-step removing, on artificially corrupted hyperspectral data *Xuzhou* (noise rate = 20%). (a) CNN. (b) CLCM. (c) CLDM. (d) CLCC. (e) CLDC. (f) ELCI. (g) ELCS. (h) ELDI. (i) ELDS.

TABLE IV
ACCURACY OF THE HARDEST CLASS IN NINE EXPERIMENTS ON ORIGINAL DATASETS

	Raw	Deletion				Correction			
		CLDM	CLDC	Margin		CLCM	CLCC	Margin	
				ELDS	ELDI			ELCS	ELCI
Indian Pines	50.00%(4)	25.00%(9)	50.00%(4)	55.00%(4)	**67.50%**(1)	35.00%(8)	47.50%(7)	60.00%(2)	55.00%(3)
PaviaU	65.19%(9)	67.12%(8)	69.57%(5)	**74.74%**(1)	71.40%(4)	67.45%(7)	73.31%(2)	68.61%(6)	72.90%(3)
Houston	42.87%(9)	51.91%(6)	46.49%(8)	**60.43%**(1)	59.47%(2)	50.85%(7)	52.66%(5)	53.29%(4)	57.87%(3)
Botswana	68.52%(9)	69.30%(7)	68.89%(8)	**87.59%**(1)	69.44%(5)	69.36%(6)	73.37%(4)	80.19%(3)	79.44%(2)
KSC	52.81%(8)	51.88%(9)	56.25%(6)	59.69%(4)	**71.25%**(1)	56.56%(5)	60.63%(3)	53.13%(7)	**71.25%**(1)
Xuzhou	68.21%(8)	69.50%(4)	68.97%(6)	70.66%(2)	68.30%(7)	69.03%(5)	66.55%(9)	**74.89%**(1)	70.33%(3)
Average	57.59%(8)	55.79%(9)	58.38%(6)	**68.02%**(1)	67.89%(2)	58.03%(7)	62.00%(5)	65.02%(4)	67.80%(3)

TABLE V
ACCURACY OF THE HARDEST CLASS IN NINE EXPERIMENTS ON ARTIFICIALLY CORRUPTED DATASETS (NOISE RATE = 20%)

	Raw	Deletion				Correction			
		CLDM	CLDC	Margin		CLCM	CLCC	Margin	
				ELDS	ELDI			ELCS	ELCI
Indian Pines	30.00%(7)	30.00%(7)	30.00%(7)	52.50%(5)	**67.50%**(1)	50.00%(4)	58.50%(2)	40.00%(6)	55.00%(3)
PaviaU	50.38%(9)	61.26%(8)	62.05%(7)	69.12(2)	69.00%(3)	63.26%(6)	**69.52%**(1)	63.50%(5)	65.69%(4)
Houston	39.26%(8)	40.74%(6)	37.13%(9)	43.72%(5)	51.06%(3)	40.32%(7)	44.36%(4)	**54.89%**(1)	**54.89%**(1)
Botswana	62.59%(7)	66.11%(5)	62.22%(8)	63.52%(6)	**78.33%**(1)	66.29%(4)	58.15%(9)	74.80%(2)	72.77%(3)
KSC	44.38%(8)	54.69%(3)	43.13%(9)	52.81%(4)	**56.88%**(1)	47.19%(7)	47.81%(6)	**56.88%**(1)	52.19%(5)
Xuzhou	57.35%(9)	68.34%(5)	65.70%(8)	71.93%(3)	70.76%(4)	68.15%(6)	67.84%(7)	**72.81%**(1)	72.03%(2)
Average	47.33%(9)	53.52%(7)	50.04%(8)	58.93%(4)	**65.59%**(1)	55.87%(5)	55.84%(6)	60.48%(3)	62.10%(2)

consistently outperforms that of nonmargin algorithms (CLDM, CLDC, CLCM, and CLCC), exceeding them by approximately 8.6 and 8.0 percentage points in the two tables, respectively, with a performance gain of around 14%. The two tables explicitly report the accuracy of the hardest class, which typically corresponds to categories with scarce samples in a long-tail distribution or those with the most indistinguishable features. Taking Indian Pines dataset in Table IV as an instance, the accuracy of CLDM is merely 25.00%, whereas ELDI reaches 67.50% and ELDS achieves 55.00%.

The experimental results demonstrate the exceptional robustness of the proposed algorithms in handling hard samples or those with ambiguous boundaries, effectively addressing the limitations where traditional algorithms fail on difficult samples. Table V also demonstrates the stability of the proposed algorithms in the presence of label noise. Even with 20% noise added, our proposed algorithms maintain high performance in scenarios where labels are both scarce and erroneous. This further confirms that the utilization of the margin mechanism effectively mitigates the impact of noise.

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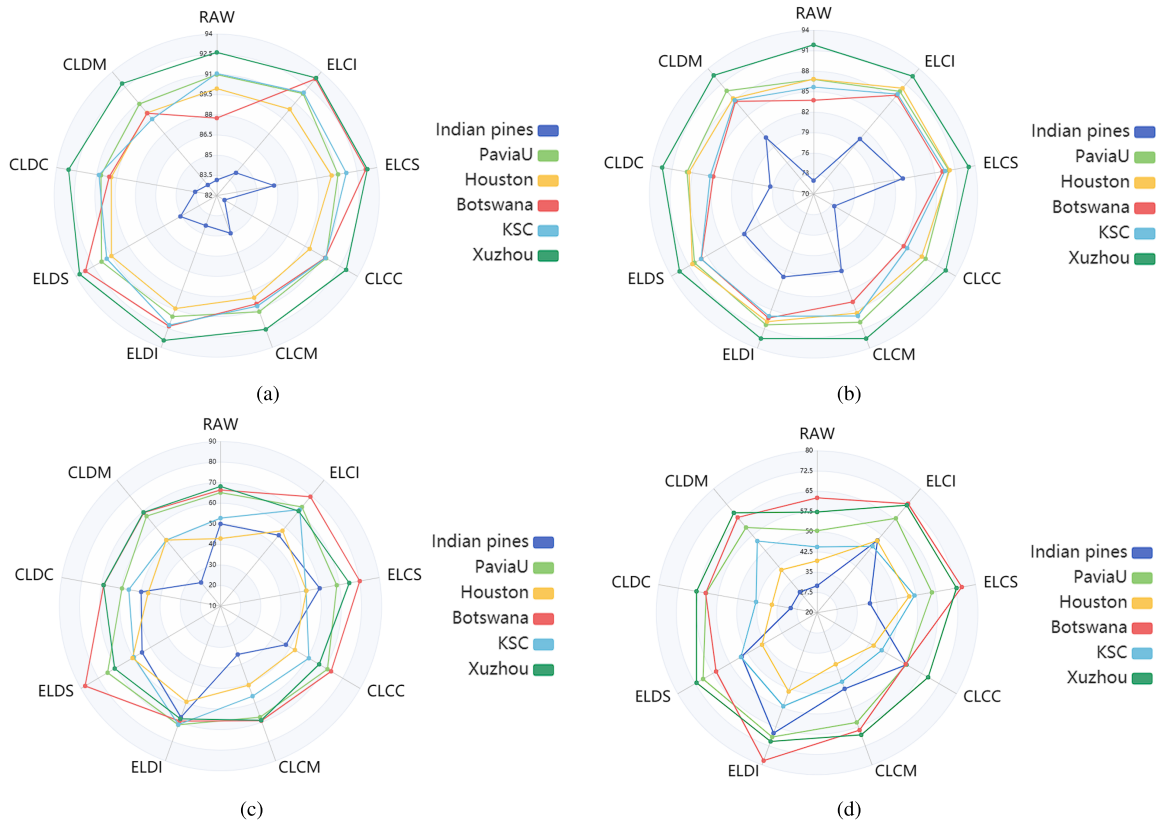


Fig. 14. Radar chart of the average values for four groups in Tables II–V. (a) OA in nine experiments on original datasets. (b) OA in nine experiments on artificially corrupted datasets (noise rate = 20%). (c) Accuracy of the hardest class in nine experiments on original datasets. (d) Accuracy of the hardest class in nine experiments on artificially corrupted datasets (noise rate = 20%).

Furthermore, the proposed method for handling class noise can additionally lead to a reduction in the computational resources, both in terms of time and space, required for CNN (or any other noise-sensitive supervised classifier), achieved by reducing the size of the training dataset through data cleaning.

3) Comprehensive Analysis: It is noteworthy that Indian Pines dataset exhibits the most substantial improvement in OA (11.27%) when the dataset is artificially corrupted by a 20% noise rate. This phenomenon can be attributed to the inherent characteristics of the dataset: a relatively small sample size coupled with complex high-dimensional spectral features. These factors render the classifier highly sensitive to class noise. Our proposed method effectively mitigates this sensitivity by robustly filtering out boundary noise, thereby stabilizing the decision boundary in data-limited scenarios. In contrast, for Xuzhou dataset, despite its very high spatial resolution and the presence of complex mixed categories that make classification challenging, the vast amount of training samples enables deep learning models to perform more robustly. Consequently, adding artificially corrupted class noise has a limited impact on accuracy, with the maximum OA only improved by 1.17%. Furthermore, the experimental results indicate that the deletion strategy sometimes outperforms correction in complex environments. This is primarily due to the risk of error propagation, which arises in two scenarios. First, when the hyperspectral labels are initially clean (original datasets), the process of correcting them actually introduces new noise. Second, in the hardest-class scenarios, accurately correcting noisy labels is extremely challenging. Incorrect correction introduces secondary noise, whereas removing ambiguous samples guarantees the purity of the remaining training set. Consequently, in both original datasets and the hardest-class scenarios, removing samples identified as noise proves to be more effective than label correction.

The following radar chart (see Fig. 14) corresponds to the accuracy in Tables II–V mentioned above, respectively.

D. McNemar's Test

Table VI shows the result of McNemar's test for statistical comparisons of classification performance of the correction iterative filter to other noise deletion and correction filters, respectively, on 20% noise rate and original datasets. The numbers in the table are Z values. If —Z— is higher than 1.96, the difference between the two classifiers is to be statistically significant. It is obvious in Table VI that most of the Z values in the first five rows are higher than 1.96, which proves the proposed algorithm exhibits significant improvement in classification performance compared to other traditional majority and consensus filters, as well as the raw deep learning classifier (CNN). In other words, this phenomenon also indicates that the proposed algorithm is more stable in uncertain noise. The last three lines are a comparison of the four EL algorithms we proposed. In contrast to other Z values above, the proposed method not only achieves the best classification performance,

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TABLE VI
MCNEMAR'S TEST AMONG NINE EXPERIMENTS ON SIX DATASETS WITH ARTIFICIALLY ADDED 0% AND 20% NOISE RATES

Noise rate	Indian Pines		PaviaU		Houston		Botswana		KSC		Xuzhou	
	0	20	0	20	0	20	0	20	0	20	0	20
CI vs raw	2.69	11.8	10.62	9.58	4.46	10.56	7.22	4.8	3.22	4.99	139.72	139.58
CI vs MC	5.92	0.89	9.54	-5.66	4.03	7.9	3.83	2.27	4.21	0.45	7.43	11.62
CI vs MD	0.47	-0.09	11.24	0.02	4.98	1.97	5.41	1.74	2.83	1.96	5.49	15.65
CI vs CC	5.04	21.36	7.68	1.53	3.92	3.44	2.75	3.32	2.01	5.77	7.71	12.86
CI vs CD	2.33	10.01	8.33	1.67	7.28	5.06	5.07	4.06	2.32	5.55	3.87	12.96
CI vs CS	1.13	-4.29	8.77	-9.27	3.3	-1.59	0	0	1.26	0.23	1.16	2.82
CI vs DS	0.66	-1.55	3.52	-3.04	3.3	-1.3	0.63	0.41	0.97	0	-3.14	12.19
CI vs DI	-9.75	-0.27	0.2	3.56	1.71	3.71	0.6	-0.79	0.09	-0.13	1.91	0.34

TABLE VII
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON UP AND KSC DATASETS (NOISE RATE = 20%)

Method / Dataset	DPNLD	K-SDP	RLPA	3D-CNN	SSRN	DCRN	ELDS	ELDI	ELCS	ELCI
PaviaU	75.05%	78.04%	87.00%	85.69%	90.12%	90.33%	90.10%	**90.37**%	90.10%	89.66%
KSC	57.59%	58.22%	56.03%	86.81%	87.53%	88.96%	88.99%	89.02%	**89.54**%	89.13%

but also has smaller data differences among them. This can be easily confirmed by the following table.

E. Comparative Experiments

To demonstrate its effectiveness, the proposed noise deletion algorithm is compared with multiple baselines on the widely used KSC and PaviaU datasets, which provide a comprehensive evaluation across distinct scene types. This experiment employs six methods as comparative baselines, including DPNLD [66], K-SDP [67], RLPA [68], 3D-CNN, SSRN [69], and DCRN [39]. Specifically, DPNLD, K-SDP, and RLPA are data cleaning methods that clean the training data by removing examples potentially affected by contaminated labels. Correspondingly, 3D-CNN, SSRN, and DCRN are deep learning methods that construct supervised learning architectures robust to noisy labels. All of the evaluation metrics for the experimental results are based on the OA. To ensure fairness, the training and testing data used in the experiment are identical. (The training data contain 20% noise.) The experimental results are shown in the following table.

As presented in Table VII, under the challenging experimental condition of a 20% label noise rate, the proposed methods (ELDS, ELDI, ELCS, and ELCI) exhibited superior noise robustness and generalization capabilities. In contrast to traditional methods (e.g., DPNLD), which suffered significant performance degradation, and mainstream deep learning models (e.g., 3D-CNN and DCRN), the proposed algorithms maintained consistent high performance. Specifically, they achieved the highest classification accuracies of 90.37% and 89.54% on the PaviaU and KSC datasets, respectively. These results strongly demonstrate the significant advantages of the proposed approach in mitigating label noise interference and extracting discriminative features from noisy data, outperforming existing comparative methods.

V. CONCLUSION

This article offers valuable insights into the behavior of supervised classifiers, informing the development of operational implementations for various hyperspectral remote sensing tasks, especially those involving complex dynamics and heterogeneous environments. This study highlights the impact of mislabeling in training data on both overall and hardest-class performances, which can further deteriorate when the classifier encounters another significant issue in the training data: the hardest class. To this end, we propose an ensemble filtering technique that enables the application of noise-sensitive filters to address the complexities of such

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challenging datasets. Relative to voting-driven ensemble filters, the proposed method improves overall and hardest-class accuracies by 1.01% and 6.02%, respectively, when tested on unmodified datasets. Furthermore, on datasets with artificial corruption, it delivers enhancements of 1.09% in OA and 9.72% in the most challenging class.

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